

Altaf Ahmed Mushtaq Ahmed

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Profile Summary

Backend Developer with 4+ years of experience in designing and building scalable, secure, and high-performance backend systems and APIs. Skilled in multiple programming languages including **Python, Go, and Java**, with additional expertise in Node.js and TypeScript using frameworks like Express.js and NestJS. Strong understanding of distributed systems, including fault tolerance, data consistency, and scalability principles. Proficient in microservices architecture, RESTful APIs, and real-time communication. Experienced in integrating relational and NoSQL databases (PostgreSQL, MySQL, MongoDB), implementing authentication (JWT, OAuth2), and optimizing system performance. Skilled in containerization (Docker), cloud deployment (AWS), and orchestration strategies. Experienced in implementing efficient caching mechanisms (Redis) and ensuring secure application development. Committed to designing resilient and high-performance backend systems that power modern distributed applications.

Technical Skills

- Programming Languages & Databases:** Python, Java, Go, JavaScript, TypeScript, SQL, Rust, Vyper, MongoDB, PostgreSQL, MySQL, Firebase, SQL-based Analytics
- Blockchain & Web3 Technologies:** Ethereum & EVM Ecosystem, Solana, Binance Smart Chain (BSC), Polygon, Hyperledger Fabric, Corda, Chaincode Development, Smart Contracts (Solidity, Rust, Vyper), NFT Standards (ERC-20, ERC-721, ERC-1155), DAO Architecture, Tokenomics, IPFS, The Graph, Chainlink Oracles, Metamask Integration, WalletConnect, Blockchain Infrastructure Management, Layer 2 Solutions (Arbitrum, Optimism)
- Frameworks, Libraries & Tools:** Hardhat, Truffle, Remix IDE, Ganache, Web3.js, Ethers.js, OpenZeppelin, Spring Boot, Django, Gin (Go), React.js, Next.js, Redux, Node.js, Express.js, RESTful APIs, HTML5, CSS3
- Cloud, DevOps & Microservices:** AWS (EC2, S3, Lambda, RDS, DynamoDB, CloudWatch, CodePipeline), Google Cloud Platform (GCP), Microsoft Azure, Docker, Kubernetes, Git, GitHub, Jenkins, CI/CD Pipelines, Distributed Systems
- Security, Testing & Development Practices:** Smart Contract Auditing (Slither, MythX, Oyente), Blockchain Security, Gas Optimization, Unit Testing (Mocha, Chai, Jest), Agile, Scrum, Kanban, Test-Driven Development (TDD), SDLC, Jira, Confluence

Professional Experience

Blockchain Developer, ChainLogic Systems

Jan 2025 – Present | USA

- Developed smart contracts using Solidity and Ethereum to enable decentralized lending, borrowing, and staking on the DeFi platform, while ensuring low transaction costs and optimizing gas fees with Layer-2 solutions like Optimism and Arbitrum.
- Engineered secure and scalable blockchain applications by integrating Web3.js for frontend interaction and leveraging Node.js and Express.js for backend development, enabling seamless communication with smart contracts.
- Maintained blockchain nodes (Ethereum and Binance Smart Chain), ensuring high availability, uptime, and system security, including performing regular blockchain security audits and using OpenZeppelin for cryptographic integrity.
- Optimized backend performance through efficient smart contract logic and automated CI/CD pipelines with Jenkins and Docker, resulting in a 30% increase in process efficiency and improved platform scalability.
- Collaborated with teams to integrate IPFS for decentralized file storage, designed API interfaces for third-party integrations, and created user-friendly dashboards using React.js to visualize blockchain transaction data.

Blockchain Developer, Web3 NFT Platform (Capstone Project)

Aug 2024 – Dec 2024 | USA

- Engineered a decentralized Web3 platform on the Ethereum blockchain using Azure Blockchain Workbench, enabling NFT minting and ownership lifecycle management via Solidity-based smart contracts adhering to the ERC-721 standard, minting over 100 NFTs with full on-chain verifiability and access control.
- Integrated IPFS and Pinata for immutable, decentralized storage of NFT metadata, achieving 99.9% system uptime and ensuring content-addressable integrity across the blockchain network.
- Developed a full-stack dApp with React, Web3.js, and REST APIs, optimizing gas estimation and reducing transaction latency by 35% while supporting seamless wallet integrations, including MetaMask.
- Built a real-time analytics dashboard using Plotly Dash and The Graph protocol to track transaction throughput, ownership events, and contract interactions, providing actionable insights and transparency to users.
- Utilized OpenZeppelin libraries for secure smart contract development, and performed extensive unit testing with Truffle and Ganache, ensuring bug-free deployments and optimal contract performance on Ethereum testnets (Rinkeby, Goerli) before mainnet deployment.

Backend Engineer, Digital Ledger Voting System (Capstone Project)

Aug 2021 – Jan 2022 | India

- Developed and deployed a secure decentralized voting application (dApp) on a private Ethereum network (Geth), enhancing voter accessibility by 40% and ensuring zero fraud incidents in controlled simulations.
- Engineered a Layer 2 scaling solution (zkRollup/Plasma) to optimize blockchain consensus, reducing gas costs by 60% and transaction latency by 50%, improving system efficiency and scalability.
- Implemented a custom identity protocol leveraging off-chain cryptographic proofs and SHA-256 voter authentication, reducing identity fraud by 90% and improving KYC throughput by 50%.

- Developed and deployed Solidity smart contracts to automate the voting process, achieving 95% process automation with tamper-proof audit trails, ensuring transparency and accountability.
- Integrated Merkle tree hashing for secure, verifiable vote recording, enabling deterministic proof validation without exposing full data, and preserving both privacy and transparency.
- Built real-time analytics dashboards with Python, SQL, and Matplotlib, providing insights on voter turnout, regional distribution, and election status, enabling data-driven decision-making.

Associate Software Engineer, Flexera Software Private Limited

July 2022 – Aug 2023 | India

- Architected cloud-native API connectors and modular ETL pipelines using Java, Python, and SQL to automate ingestion and processing of external software vulnerability feeds; deployed on AWS EC2 and orchestrated using AWS Step Functions.
- Automated SBOM (Software Bill of Materials) scanning workflows with AWS Lambda and AWS Batch, cutting manual intervention by 40% and streamlining vulnerability detection across distributed codebases.
- Designed and deployed real-time analytics dashboards with Power BI, Python, and Amazon QuickSight, integrated via Amazon RDS, enabling executive-level visibility into vulnerability trends across CI/CD pipelines.
- Led CI/CD automation using AWS CodePipeline, JUnit, and Maven, reducing release cycle times by 30% and boosting deployment reliability through rigorous automated testing and version control practices.
- Built and deployed containerized microservices with Docker, managed via Amazon ECS, and implemented comprehensive monitoring and logging via Amazon CloudWatch Logs to enhance operational visibility and incident response times.
- Collaborated within Agile development teams, participated in daily stand-ups, sprint planning, and code reviews, while promoting DevOps best practices and ensuring high-quality code delivery through unit testing and peer feedback.

Software Engineer, Cogknit Semantics Pvt. Ltd

Oct 2021 – Apr 2022 | India

- Built and deployed user behavior analytics using Python, Pandas, and Matplotlib, processing large datasets stored in Amazon S3 and transformed via AWS Glue, leading to a 30% increase in user satisfaction through actionable insights into user preferences and behaviors.
- Integrated a personalized recommendation system using Amazon Personalize, leveraging backend logic and AWS Lambda for dynamic content delivery. This system increased user engagement by 35% and reduced user drop-off by 20%.
- Engineered a scalable log processing pipeline using AWS Lambda, Amazon CloudWatch, and EventBridge to generate real-time KPIs and alert systems, improving leadership decision-making and reducing response times for issues.
- Automated SQL-based reporting and dashboard updates using AWS Athena and CloudWatch Events, reducing manual effort and ensuring data-driven insights were delivered to stakeholders efficiently.

Leadership Experience

Vice president ISACA YU Student Club, ISACA New York Chapter

Aug 2024 – May 2025 | New York, USA

- Managed a **15-member team**, delegating tasks, resolving conflicts, and ensuring smooth execution of all club activities.
- Organized **5+ cybersecurity hackathons and workshops**, driving participation from students and industry professionals.
- Hosted **networking sessions and technical events** to enhance member engagement and foster connections with ISACA New York Chapter leaders.
- Coordinated **logistics, sponsorships, and communications** for events, improving attendance and overall member experience.

Educations

Master of Science, Yeshiva University

Aug 2023 – May 2025 | New York, USA

Cyber Security

Bachelor of Technology, PES University

July 2018 – May 2022 | Bangalore, India

Computer Science and Engineering

Certificates

Certified Blockchain Developer – Blockchain Council

Credential ID: 134966407 | Issued: 2025

Publication

IEEE Publication

Nov 2022

IEEE Research Paper: Intelligent Digital Ledger Voting System Published in 2022 4th International Conference on Advances in Computing, Communication Control and Networking (ICAC3N)